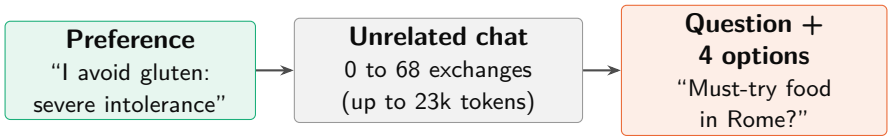
Improving Long-Context Preference Following in LLMs

via a Trained Preference-Detection Classifier and Enhanced Retrieval

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The problem: LLMs forget what you told them



PrefEval (ICLR 2025)

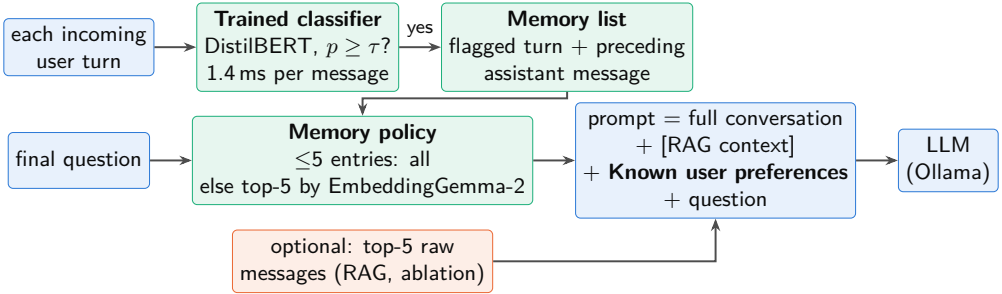
- 1,000 preference–question pairs, 20 topics
- Exactly one option respects the preference
- *Explicit*: stated directly. *Implicit*: revealed by choosing among options

Reproduced locally

- Llama zero-shot: 94.2% at 0.2k → 43.3% at 3k
- Mistral zero-shot: 95.5% → 37.6% at 23k
- Paper's best method (RAG), Mistral 23k: 53.7%

Goal: keep following preferences in long conversations, with open 7–8B models on one consumer GPU (RTX 3060, 12 GB).

Proposed pipeline, and what is new



What is new compared with the baselines

- All 5 baselines re-read the conversation at question time and keep **no state**; none has a trained component
- A **cheap trained gate** decides what to store: one forward pass, no extra LLM call
- Modern retriever benchmarked against the paper's 2021 SimCSE, and tested *with and without* memory

Phase 1: local reproduction of the baselines



- Original prompts kept; only the inference call replaced (Bedrock → Ollama)
- 64,535 questions, about 71 GPU-hours
- 12 of 29 cells within 5 points of the paper, 20 within 10
- Fixed: double start token, whitespace bug, silent truncation, broken RAG files

Trends reproduce. Largest gap: Mistral RAG at long context (53.7% vs 75%). The 8-bit KV cache is a plausible cause, not yet verified.

Phase 2: how the training data was built

12,950 user messages

Source	Label origin	Label 1	Label 0	Rows
Explicit preference	gold	988	0	988
Persona preference field	gold	30	0	30
Choice selection (implicit)	gold	991	0	991
Persona dialogue turn	pseudo (LLM)	1,020	3,355	4,375
Choice initial query	gold	0	577	577
Final question	gold	0	994	994
OpenAssistant message	pseudo (LLM)	114	4,881	4,995
Total		3,143	9,807	12,950

Manual audit of pseudo-labels

Audit stratum	n	Before refinement	After refinement
Persona, LLM-picked turn (+)	40	1.000	1.000
Persona, other turns (-)	39	0.821	0.966
Persona, LLM vs overlap disagree	40	0.475	0.909
OpenAssistant, LLM yes (+)	40	0.400	0.675
OpenAssistant, LLM no (-)	40	1.000	1.000
All strata	199	0.739 (κ=0.48)	0.907

- **Gold** labels from PrefEval's structure; **pseudo**-labels from Llama-3-8B for persona turns and OpenAssistant
- Audit: OpenAssistant "yes" labels only 40% correct (task requirements mistaken for preferences)
- Fixes: stricter re-check (174 rejected rows kept as hard negatives); union of LLM and word-overlap picks

The audit caught systematic label errors before training. Agreement: 73.9% $\rightarrow$ 90.7%.

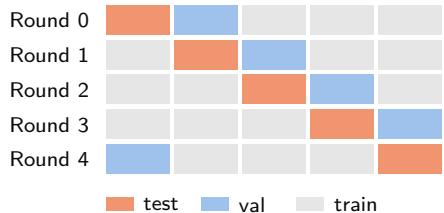
Phase 2: how the classifiers were trained

Transfer learning

- Pretrained body + new 2-class head; input = message text only
- Cross-entropy loss, AdamW, warmup then linear decay
- Keep the checkpoint with best validation F1; early stopping

	DistilBERT, ModernBERT	TinyLlama, Qwen3
Training	full fine-tuning	LoRA (r=16) + head
Trainable	100%	about 1%
LR	3e-5, 5e-5	2e-4
Epochs, batch	4, 32	2, 16×2

Leakage control: topic 5-fold CV



- 4 topics per fold; test topics unseen in training
- LMSYS distractors never trained on

4 models × 5 folds = 20 trainings, 4.3 hours in total. Both leakage rules are asserted in code.

Phase 2: classifier results

Model	F1	Prec.	Rec.	F1 std	AUC	ECE	OOD flags	ms/msg	Trainable	Train min
DistilBERT	0.969	0.983	0.955	0.007	0.994	0.008	0.4%	1.4	67M	14
ModernBERT	0.972	0.988	0.956	0.004	0.995	0.009	0.2%	6.5	150M	32
TinyLlama-1.1B (LoRA)	0.974	0.988	0.961	0.006	0.997	0.010	0.1%	17.1	13M	91
Qwen3-1.7B (LoRA)	0.976	0.988	0.965	0.006	0.997	0.009	1.5%	25.1	17M	120

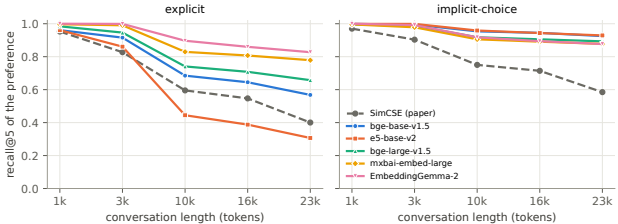
vs prompted LLM (same 2,000 rows)

Detector	F1	Prec.	Rec.	ms/msg
Prompted Llama-3-8B	0.879	0.787	0.996	72
DistilBERT	**0.990**	0.991	0.990	**1.4**

- All four within 0.7 F1 points; Qwen3 is best, but 18× slower
- Best of 33 ensembles: 0.977, only +0.001, because errors are correlated (0.68–0.75)
- Limitation: casual real-chat recall only 16–26%

DistilBERT selected: 51× faster than the LLM, more accurate, and low false alarms on the distractor turns.

Phase 3, Step 1: which encoder finds the preference?



- SimCSE at 23k: 40.0% → EmbeddingGemma-2: 82.7% (explicit)
- SimCSE ranks the preference first only 8.5% of the time at 1k
- e5-base is *worse* than SimCSE on long explicit (30.7%)
- Nobody ranks an implicit selection first → store it *with* its options

		Explicit					Implicit-choice				
Retriever	Params	R@5 1k	3k	10k	23k	R@1 23k	R@5 1k	3k	10k	23k	ms/msg
SimCSE (paper)	355M	0.95	0.83	0.60	0.40	0.05	0.97	0.90	0.75	0.58	–
bge-base-v1.5	110M	0.96	0.92	0.68	0.57	0.29	1.00	0.99	0.95	0.92	3.5
e5-base-v2	110M	0.96	0.86	0.44	0.31	0.12	1.00	1.00	0.96	0.93	3.6
bge-large-v1.5	335M	0.98	0.95	0.74	0.66	0.43	1.00	0.99	0.92	0.89	9.8
mxbai-embed-large	335M	1.00	0.99	0.83	0.78	0.53	1.00	0.98	0.91	0.88	9.8
EmbeddingGemma-2	744M	1.00	1.00	0.90	0.83	0.53	1.00	0.99	0.92	0.88	15.6

Phase 3, Step 2: the preference memory

How it works

- Score each user message with a model that never saw its topic
- Store flagged turn + preceding assistant message

Quality

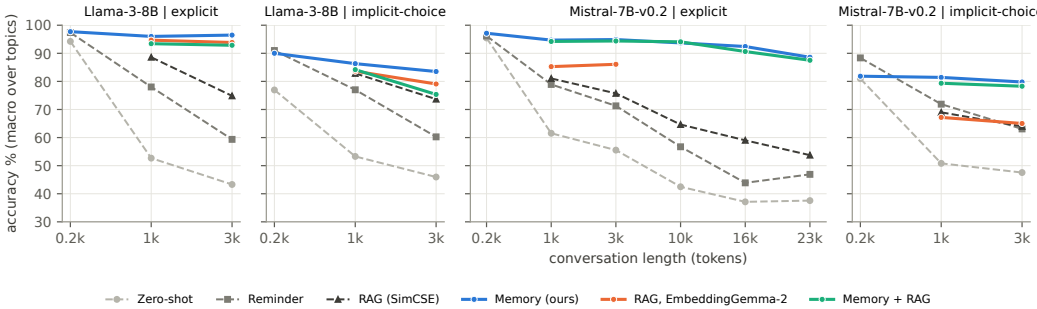
- Preference caught: **98.6%** explicit, **99.9%** implicit
- False entries: none up to 10k, 0.65 per question at 23k
- Adds about 61 prompt tokens (Llama, explicit, 3k)

What the LLM sees (implicit, shortened)

```
Known user preferences:  these are the
preferences I have stated before.
#Start of Known Preferences#
[Preference 1, turn 2]
Assistant:  There are many great hotel options
in Las Vegas...
User:  I think I'll go with The Westin.  The
others seem too lively.
#End of Known Preferences#
Please follow these preferences wherever they
are relevant to my question.
<question + 4 options>
```

Design rule: each new method changes one thing from a baseline (96 prompts verified byte-for-byte against Phase 1).

Main result: memory stays nearly flat as context grows



Mistral at 23k: 88.6% vs 53.7% for the paper's RAG, 46.9% for Reminder, 37.6% for zero-shot. From 0.2k to 23k, zero-shot and Reminder lose 49–58 points; memory loses 8.5.

Main result: the numbers

Model	Form	Length	Best baseline	Memory	Memory + RAG	Gain
Llama-3-8B	explicit	3k	74.8 (RAG)	**96.5**	92.9	+21.6
Llama-3-8B	implicit-choice	3k	73.7 (RAG)	**83.5**	75.4	+9.8
Mistral-7B-v0.2	explicit	3k	75.7 (RAG)	**94.9**	94.4	+19.2
Mistral-7B-v0.2	explicit	10k	64.6 (RAG)	93.7	**94.1**	+29.1
Mistral-7B-v0.2	explicit	16k	59.0 (RAG)	**92.4**	90.6	+33.4
Mistral-7B-v0.2	explicit	23k	53.7 (RAG)	**88.6**	87.5	+34.9
Mistral-7B-v0.2	implicit-choice	3k	64.0 (RAG)	**79.8**	78.3	+15.8

- Same questions, option order and strict metric as baselines; 46,000 questions, about 44 GPU-hours
- Beats every baseline in all 11 cells from 1k up; best overall in 10 (Mistral 10k: Memory + RAG is 0.4 higher)
- Also above the paper's published RAG: 80% for Llama at 3k, 75% for Mistral at 23k

Exception: at 0.2k there is nothing to forget. Memory ties Reminder on explicit, and loses on Mistral implicit (81.9% vs 88.4%).

Ablation 1: swapping only the retriever

Model	Form	Length	SimCSE RAG	EmbeddingGemma-2 RAG	Δ	Memory + RAG
Llama-3-8B	explicit	1k	88.6	94.6	+6.0	93.2
Llama-3-8B	explicit	3k	74.8	93.9	+19.1	92.7
Llama-3-8B	implicit-choice	1k	82.8	84.1	+1.2	84.5
Llama-3-8B	implicit-choice	3k	73.7	79.6	+6.0	75.8
Mistral-7B-v0.2	explicit	1k	81.0	85.5	+4.5	94.2
Mistral-7B-v0.2	explicit	3k	75.7	86.3	+10.6	94.5
Mistral-7B-v0.2	implicit-choice	1k	69.0	67.6	-1.4	79.3
Mistral-7B-v0.2	implicit-choice	3k	64.0	65.6	+1.6	78.2

- Same prompt, same questions: only the encoder changes
- **Explicit improves strongly:** Llama at 3k goes from 74.8 to 93.9
- **Implicit barely changes,** although retrieval itself improved: fragments of a dialogue are hard to use

A better encoder fixes the retrieval bottleneck for explicit preferences. Implicit needs the full options + selection context that memory provides.

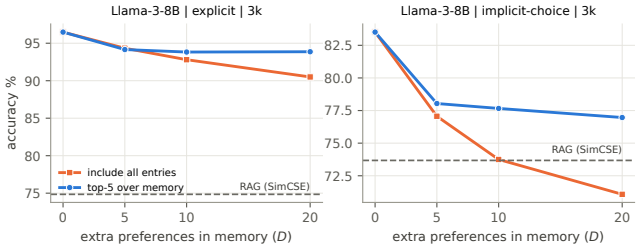
Ablation 2: adding RAG on top of memory does not help

Model	Form	Length	Lost	of which format	wrong option	pref. retrieved	Gained	Same answer %
Llama-3-8B	explicit	3k	49	40	9	49	11	93.8
Llama-3-8B	implicit-choice	3k	99	65	34	98	16	85.2
Mistral-7B-v0.2	explicit	3k	14	0	14	14	7	97.5
Mistral-7B-v0.2	explicit	23k	28	0	28	22	16	95.1
Mistral-7B-v0.2	implicit-choice	3k	46	1	45	46	26	90.0

- **Llama:** mostly format failures. It writes "Based on your previous..." and the 5-token limit cuts it off (65 of 99 implicit losses)
- **Retrieval did not fail:** the preference was retrieved in 98 of those 99
- **Mistral:** its prompt pre-fills `<choice>`, so no format failures; same answer on 90–98% of questions

With about 99% classifier recall, there is little left for raw-message retrieval to recover. It mainly adds noise.

Ablation 3: what if memory holds many preferences?



Form	D	Include all entries	Include all accuracy	Top-5 retrieval over memory pref. kept (%)	Top-5 retrieval over memory accuracy
explicit	0	1	96.5	100.0	96.5
	5	6	**94.3**	100.0	94.2
	10	11	92.8	99.1	**93.8**
	20	21	90.5	97.4	**93.9**
implicit-choice	0	1	83.5	100.0	83.5
	5	6	77.1	100.0	**78.0**
	10	11	73.7	99.9	**77.7**
	20	21	71.1	98.5	**77.0**

- Added D preferences from *other* topics, in random order (Llama, 3k tokens)
- Top-5 retrieval keeps the true preference for 97.4% / 98.5% of questions at $D{=}20$

Retrieval is most useful over the memory: at $D{=}20$, include-all falls to 90.5% / 71.1%; top-5 holds 93.9% / 77.0%.